

SAIES Control Tower

Data Dictionary

NSF Undergraduate Research Program · CSULA · Spring/Summer 2026

Key types	PK Primary Key — uniquely identifies each row	FK Foreign Key — links to another table	— Measure or attribute column
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FACT TABLE	fact_sales
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The core transaction record. One row per sales order line item — captures what was sold, to whom, where, when, at what price, and how much profit it generated.

Technical name	Plain English meaning	Data type	Key	Example value
fact_sales_id	Unique record ID	INT	PK	1, 2, 3...
product_key	Links to product catalog	INT	FK	7 → Cheese
location_key	Links to region and city	INT	FK	3 → Hamburg, North
customer_key	Links to customer account	INT	FK	5 → Customer 80295
sales_org_key	Links to sales organization	INT	FK	2 → Org O3
date_key	Links to calendar date	INT	FK	12 → Jan 14, 2022
sim_key	Links to simulation round and step	INT	FK	8 → Round 2, Step 5
SALES_ORDER_NUMBER	Order number	INT	—	3241
LINE_ITEM	Line position within the order	INT	—	1, 2, 3
QUANTITY	Units ordered	INT	—	39
QUANTITY_DELIVERED	Units actually delivered	INT	—	39
NET_PRICE	Price per unit (EUR)	DECIMAL	—	26.99
NET_VALUE	Total sale amount (EUR)	DECIMAL	—	1,052.61
COST	Total cost of goods sold (EUR)	DECIMAL	—	900.90

Technical name	Plain English meaning	Data type	Key	Example value
CONTRIBUTION_MARGIN	Profit on this sale (EUR)	DECIMAL	—	151.71
CONTRIBUTION_MARGIN_PCT	Profit as a percentage of revenue	DECIMAL	—	0.14 = 14%

FACT TABLE	fact_pricing
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Price history record. One row per product price setting — tracks how each sales organization changed prices across simulation rounds.

Technical name	Plain English meaning	Data type	Key	Example value
fact_pricing_id	Unique record ID	INT	PK	1, 2, 3...
product_key	Links to product catalog	INT	FK	3 → Milk
sales_org_key	Links to sales organization	INT	FK	1 → Org O3
date_key	Links to calendar date	INT	FK	1 → Jan 1, 2022
sim_key	Links to simulation round and step	INT	FK	3 → Round 1, Step 3
DISTRIBUTION_CHANNEL	Sales channel code	INT	—	16
DC_NAME	Sales channel name	VARCHAR	—	Retailers
PRICE	Listed price in EUR	DECIMAL	—	26.99
CURRENCY	Currency used	VARCHAR	—	EUR

DIMENSION TABLE	dim_product
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Product catalog. Describes each of the 6 dairy products sold across all 7 sales organizations (42 unique product combinations).

Technical name	Plain English meaning	Data type	Key	Example value
product_key	Unique product ID	INT	PK	1 through 42
MATERIAL_NUMBER	SAP product code	VARCHAR	—	OO-T01
MATERIAL_DESCRIPTION	Product name	VARCHAR	—	Milk

Technical name	Plain English meaning	Data type	Key	Example value
MATERIAL_TYPE	Product category	VARCHAR	—	Trading Goods
MATERIAL_CODE	Short product code	VARCHAR	—	T01
MATERIAL_LABEL	Display label	VARCHAR	—	Milk

DIMENSION
TABLE

dim_location

Geographic information. Describes where orders were shipped and which warehouse fulfilled them across 3 areas (North, West, South) and 9 German regions.

Technical name	Plain English meaning	Data type	Key	Example value
location_key	Unique location ID	INT	PK	1 through 12
STORAGE_LOCATION	Warehouse code	VARCHAR	—	03N = North warehouse
REGION	German state or region	VARCHAR	—	Hamburg
AREA	Sales area	VARCHAR	—	North, West, or South
CITY	Delivery city	VARCHAR	—	Hamburg
COUNTRY	Country	VARCHAR	—	Germany
POSTAL_CODE	Postal code	INT	—	22769

DIMENSION
TABLE

dim_customer

Customer accounts. The 12 retail customer accounts that placed orders during the simulation.

Technical name	Plain English meaning	Data type	Key	Example value
customer_key	Unique customer ID	INT	PK	1 through 12
CUSTOMER_NUMBER	Customer account number	INT	—	80295
DISTRIBUTION_CHANNEL	How the customer buys (channel code)	INT	—	16 = Retailers

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DIMENSION TABLE

dim_sales_org

Sales organizations. The 7 competing teams in the ERPSim simulation, each managing their own pricing and distribution strategy.

Technical name	Plain English meaning	Data type	Key	Example value
sales_org_key	Unique org ID	INT	PK	1 through 7
SALES_ORGANIZATION	Team or organization code	VARCHAR	—	O3, P3, Q3, R3, S3, T3, U3

DIMENSION TABLE

dim_date

Calendar dates. Covers 41 unique dates from January 1 to February 10, 2022, spanning the full simulation period.

Technical name	Plain English meaning	Data type	Key	Example value
date_key	Unique date ID	INT	PK	1 through 41
SIM_CALENDAR_DATE	Full calendar date	DATE	—	2022-01-03
SIM_DATE	Simulation date label	VARCHAR	—	01/03
SIM_PERIOD	Billing or reporting period number	INT	—	1 = January, 2 = February

FACT TABLE

fact_carbon

Carbon emissions record. One row per emission event — tracks CO2e generated by goods movements, overstock, purchased products, and overhead across all 7 sales organizations.

Technical name	Plain English meaning	Data type	Key	Example value
fact_carbon_id	Unique record ID	INT	PK	1, 2, 3...
sales_org_key	Links to sales organization	INT	FK	2 → Org O3
date_key	Links to calendar date	INT	FK	1 → Jan 1, 2022
sim_key	Links to simulation round and step	INT	FK	3 → Round 1, Step 3

Technical name	Plain English meaning	Data type	Key	Example value
ROW_ID	Original row number from source file	INT	—	1, 2, 3...
TYPE	Emission category	VARCHAR	—	Goods Movement, Overstock, Overhead, Products Purchased
SUBTYPE	Specific activity within the category	VARCHAR	—	Goods Receipt, Deliveries, Internal Transfers
ORIGIN	Source warehouse or vendor code	VARCHAR	—	V04 = vendor, 03N = North warehouse
DESTINATION	Destination warehouse code	VARCHAR	—	03N, 03W, 03S
PRODUCT	Product code tied to the emission	VARCHAR	—	T01 = Milk, T02 = Cream...
MATERIAL_DOCUMENT	SAP document reference number	BIGINT	—	5000000178
SCOPE	GHG Protocol emission scope	INT	—	1 = Direct, 2 = Indirect energy, 3 = Value chain
EMISSIONS	Activity-level CO2e for this event	DECIMAL	—	100
TOTAL_CO2E_EMISSIONS	Total CO2e including upstream effects	DECIMAL	—	421,694 across all rows

DIMENSION TABLE

dim_simulation

ERPsim game structure. Tracks the 4 simulation rounds and their individual steps, enabling round-over-round performance comparison.

Technical name	Plain English meaning	Data type	Key	Example value
sim_key	Unique simulation step ID	INT	PK	1 through 41
SIM_ROUND	Game round number	INT	—	1, 2, 3, or 4
SIM_STEP	Step number within the round	INT	—	1 through 10

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Technical name	Plain English meaning	Data type	Key	Example value
SIM_ELAPSED_STEPS	Total steps completed so far	INT	—	1, 2... up to 40